

HAIL: Bridging AI, HCI, and Learning Sciences for Human-AI Interaction in Learning

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AI-supported learning sits at the intersection of AI capabilities, interaction design, learning processes, and educational practice, yet interdisciplinary collaboration can be challenging because communities may differ in how they frame problems, select methods, evaluate evidence, and involve educational stakeholders. Human-AI Interaction for Learning (HAIL) brings together researchers and practitioners across AI, human-computer interaction, the learning sciences, and educational practice to examine how such collaborations are formed, negotiated, and sustained. Building on our 2026 workshop at the International Conference on Artificial Intelligence in Education (AIED), this CHI edition shifts attention from primarily discussing AI-supported learning systems to the collaboration processes that make such work possible. Through thematic presentations, structured cross-community discussions, and small-group activities, participants will examine recurring collaboration challenges and promising practices. The workshop will produce a participant-informed synthesis of these challenges, practices, and open questions for future interdisciplinary work.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**.

Additional Key Words and Phrases: human-AI interaction, learning, interdisciplinary collaboration, collaboration practices, research communities

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1 Motivation and Goals

AI-supported learning sits at the intersection of AI capabilities, interaction design, learning processes, and educational practice. This intersection is reflected in work on real-time classroom orchestration [11], AI-assisted learning analytics for collaborative programming [10], the fit between emerging interactive technologies and educators' pedagogical

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practices [5], and student-centered evaluations of generative AI feedback [12]. Yet interdisciplinary collaboration remains challenging: collaborators may differ in how they frame problems, select methods, evaluate evidence, and involve educational stakeholders. HAIL therefore focuses on how interdisciplinary collaborators negotiate shared goals, methods, evidence, and responsibilities in human–AI learning research.

Existing scholarship offers important foundations for addressing these challenges. Co-design and participatory design foreground the roles of researchers, designers, and stakeholders in shaping technology [2, 8], while design-based implementation research emphasizes iterative collaboration around persistent problems of practice [7]. Work on teacher–AI complementarity, human–GenAI interaction, and boundary objects further illustrates how technical capabilities, pedagogical knowledge, and differing perspectives can be coordinated across communities [4, 6, 9]. Together, these traditions provide useful foundations for examining the practices and tensions of interdisciplinary collaboration in human–AI learning research.

These questions are particularly relevant to CHI because they concern not only what human–AI learning systems should be designed, but also how human-centered research on such systems is conducted across disciplinary and professional boundaries. CHI 2027’s theme of *bridges* explicitly connects disciplines and research with practice [1]. HAIL provides a focused setting for examining how these bridges are built, where collaboration breaks down, and what practices and infrastructures can support sustained interdisciplinary work.

We organize the workshop around four guiding questions:

- (1) **How can collaborators develop shared research problems across AI, HCI, and the learning sciences?** How can technical opportunities, human needs, and learning goals be negotiated into questions that different communities can jointly pursue?
- (2) **How can collaborators integrate methods and forms of evidence across communities?** Where do computational, interaction, and educational approaches complement one another, and where do their assumptions or standards of evidence differ?
- (3) **How can educational practitioners and learners meaningfully shape interdisciplinary research?** How can their situated knowledge inform problem formulation, system design, and evaluation?
- (4) **What enables interdisciplinary collaborations to persist beyond individual projects?** What practices, shared resources, infrastructures, and community structures support sustained collaboration?

The workshop aims to identify recurring collaboration challenges, compare practical approaches for negotiating goals, methods, and evidence, and surface shared questions and opportunities for future collaboration.

2 Previous Edition and Lessons for CHI

Our first workshop, *Designing and Evaluating Next-Generation Learning Interfaces: Linking AI, HCI, and the Learning Sciences*, took place at AIED 2026 and attracted 62 participants, with 19 accepted contributions [3]. Its panel, presentations, demonstrations, and community discussion showed substantial interest in connecting perspectives across AI, HCI, and the learning sciences and established a community and organizational foundation for this continuation.

The AIED edition also revealed a gap that motivates the CHI workshop. Discussions often centered on educational AI systems and research methods, while the processes through which interdisciplinary collaborations are formed and sustained remained less explicit. Participants surfaced differences in terminology, problem framing, and standards of evidence, as well as disciplinary silos and the need for stronger participation from educational practice. The CHI edition therefore shifts the focus from primarily discussing AI-supported learning systems to examining the collaboration

processes that make such interdisciplinary work possible. These observations directly informed the workshop’s guiding questions, participant recruitment strategy, and emphasis on structured cross-community discussion and synthesis.

3 Participants and Contributions

We aim to welcome approximately 40–60 participants, subject to venue capacity, including researchers, designers, teachers, students, and educational practitioners. We expect to accept 12–16 position papers, with quality and thematic coverage determining the final number. Accepted-paper authors will receive priority in accordance with CHI procedures, while the workshop will remain open to other conference attendees.

3.1 Three Directions for Position Paper Submissions

We invite accessible, non-anonymous position papers of one to two pages, excluding references. Submissions should address collaboration across AI, HCI, and the learning sciences in research on human–AI interaction for learning, including educational practitioners and other stakeholders. Authors should identify the communities and stakeholders involved and explain what their case or perspective reveals about interdisciplinary collaboration. Submissions may contribute through one or more of the following directions:

- (1) **Collaboration experiences and lessons.** Describe a concrete interdisciplinary collaboration, including who participated, how decisions were negotiated across communities, what worked or proved difficult, and what others might learn. Unsuccessful and unfinished collaborations are welcome.
- (2) **Collaboration questions and needs.** Introduce a research or practice problem that would benefit from cross-community collaboration. Explain why collaboration is needed, what expertise is available or missing, and what knowledge, methods, resources, or partnerships would help move the work forward. Prior collaboration experience is not required.
- (3) **Perspectives from educational practice.** Reflect on how educators, learners, or other educational stakeholders participate in interdisciplinary collaborations. Contributions may discuss practical needs, barriers, opportunities, or ways stakeholders can shape research questions, technologies, methods, and interpretations.

Each submission should identify a concrete experience, challenge, or problem; its implications for interdisciplinary collaboration; and one or more questions for workshop discussion. New empirical results are optional.

3.2 Recruitment, Selection, and Preparation

We will circulate the call through the previous workshop community and relevant HCI, AIED, learning sciences, learning analytics, and computing education networks. Recruitment will seek complementary disciplinary backgrounds, career stages, and collaboration experiences while explicitly welcoming perspectives from educational practice.

Organizers will review submissions for relevance, specificity, discussion potential, and complementary perspectives. Practitioner knowledge and experience will be considered alongside academic contributions. Selection will therefore consider not only the strength of individual submissions, but also how the overall set of participants can contribute complementary experiences and perspectives to the workshop discussions.

To support participant selection and group formation, an online form will collect participants’ disciplinary backgrounds, collaboration experiences, expertise they can contribute, and challenges or questions they hope to discuss. Responses will help us balance perspectives across AI, HCI, the learning sciences, and educational practice and form groups around complementary expertise and shared collaboration challenges.

Table 1. Workshop flow across two 90-minute sessions.

Session 1: From cases to recurring collaboration tensions	
Framing and cases	Introduce the workshop goals and lessons from AIED 2026, and present concrete collaboration experiences and needs.
Identify tensions	Compare cases across thematic batches to surface recurring differences in goals, assumptions, methods, evidence, expertise, and stakeholder involvement. <i>Outcome: a small set of shared collaboration tensions.</i>
Session 2: From tensions to ways forward	
Work through tensions	Mixed groups examine one tension from Session 1, identifying its context, relevant perspectives, and possible practices, resources, or infrastructures. <i>Outcome: a collaboration map.</i>
Cross-group synthesis	Compare maps to identify recurring practices, disagreements, open questions, and opportunities for continued collaboration. <i>Outcome: a shared synthesis and next steps.</i>

We propose a submission deadline of February 11, 2027; notifications will precede CHI’s early registration deadline by at least seven days. Contributors will provide a brief introduction, a discussion question, and grouping interests. With permission, accepted papers will be available on the workshop website before the workshop. The website and submission form are [website] and [submission-form], respectively.

4 Workshop Activities

The workshop will take place in person across two consecutive 90-minute sessions, separated by the conference break. The two sessions serve distinct but connected purposes. Session 1 is diagnostic: participants use concrete collaboration experiences to surface and compare recurring tensions across communities. Session 2 is constructive: mixed groups take these tensions as starting points and examine what perspectives, practices, resources, or infrastructures might help collaborators work through them. In this way, the outputs of Session 1 directly structure the group discussions and synthesis in Session 2.

4.1 Brief Presentations and Identifying Shared Tensions

Session 1 will begin with a brief introduction and lessons from AIED 2026. Each accepted position-paper contribution will receive approximately two minutes to introduce its context, collaboration experience or need, and a question for discussion. Contributions will be organized into three or four thematic batches based on recurring collaboration questions rather than discipline.

After each batch, facilitators will help participants compare cases and identify recurring collaboration tensions. Discussion will focus on questions such as: Where do assumptions or goals differ across communities? What expertise or stakeholder perspectives are missing? Where do methods or standards of evidence conflict? By the end of Session 1, organizers and participants will consolidate these observations into a small set of shared tensions that will structure the second session. During the conference break, organizers will finalize these tensions and group assignments, using participants’ stated interests and expertise to ensure complementary perspectives within each group.

4.2 Working Through Collaboration Tensions

Session 2 will shift from identifying recurring tensions to examining how they might be addressed. To reduce reorganization time, provisional groupings will be prepared in advance from submission materials and participant forms, then adjusted during the break based on the tensions surfaced in Session 1. Participants will form approximately six to eight

mixed groups, each centered on one tension surfaced in Session 1. Groups will intentionally combine complementary perspectives rather than participants from a single discipline or submission category. For example, an AI researcher, HCI researcher, learning scientist, and educational practitioner may examine the same tension from different perspectives.

A facilitator will guide each group in examining why the tension arises, which perspectives and stakeholders are involved, and what practices, resources, or infrastructures might help collaborators navigate it. Each group will create a concise *collaboration map* capturing: (1) the central tension and its context; (2) relevant perspectives, expertise, and stakeholders; (3) underlying differences or unresolved questions; and (4) promising practices, resources, or future opportunities. Groups will document their maps using a shared template and then compare them during the cross-group synthesis. These materials will inform the post-workshop synthesis.

5 Access, Documentation, and Follow-Up

We will share materials in accessible digital formats in advance, support paper or digital note-taking, and provide both written and spoken ways to contribute. Organizers will ask about access requirements ahead of time and coordinate requests with CHI.

Within four weeks after the workshop, we will circulate a draft synthesis of collaboration challenges, promising practices, and open questions for participant comment. The synthesis will organize insights around shared problem framing, methods and evidence, stakeholder involvement, and resources for sustaining collaboration, while preserving disagreements and contextual differences. Rather than forcing consensus, it will document where perspectives converge, diverge, or depend on context. A revised version will be posted on the workshop website. With authors' permission, accepted position papers will also be posted there as non-archival materials. The workshop website will also serve as a lightweight hub for sharing participant resources and supporting follow-up connections after CHI.

Workshop notes will support this collaborative synthesis. We will seek permission before attributing statements or sharing identifiable examples, and discussions will not be recorded by default. Any subsequent use of workshop materials for research publication will be subject to appropriate ethics review and consent requirements.

6 Organizers

Our organizing team spans AI, HCI, learning sciences, computing education, learning analytics, and educational practice. Collectively, the organizers have developed and studied AI-supported learning systems, worked with educators in authentic settings, and led workshops, conferences, and research communities across disciplinary boundaries. The team itself reflects the cross-community perspective that the workshop seeks to foster, bringing together expertise in technical development, interaction design, learning research, and educational practice. This complementary expertise supports participant recruitment, cross-community facilitation, and post-workshop synthesis.

Meng Xia is an Assistant Professor of Computer Science and Engineering at Texas A&M University. Her research focuses on human-AI interaction, visualization, and AI-supported learning, including classroom orchestration. She co-organized the AIED 2026 edition and will coordinate the workshop and final synthesis.

Yan Chen is an Assistant Professor of Computer Science at Virginia Tech. His research includes programming support tools, real-time learning analytics, learning at scale, and interactive AI systems for education. He will coordinate participant information collection and form design.

Qiao Jin is an Assistant Professor of Computer Science at North Carolina State University. Her research explores XR- and AI-driven technologies for teaching, learning, and social interaction in educational settings. She brings prior

CHI workshop and CSCW community-organizing experience and will coordinate the workshop website and online materials.

Yang Shi is an Assistant Professor of Computer Science at Utah State University. His research focuses on intelligent tutoring systems, student modeling, educational data mining, and machine learning for computing education. He has co-organized the CSEDM workshop series and will coordinate submission review and group formation.

Paul Denny is an ACM Distinguished Member and Professor at the University of Auckland. His research centers on collaborative learning and computing education, and he created PeerWise, a large-scale student-generated learning platform. He will support outreach and cross-community discussion.

Tiffany Barnes is a Distinguished Professor of Computer Science at North Carolina State University. Her research spans computing education, educational data mining, AI-supported learning, and inclusive computing education. She will co-lead outreach and community engagement.

Qingsong Wen is Head of AI and Chief Scientist at Squirrel Ai Learning and a PhD Supervisor at the University of Oxford. His research spans machine learning, AI for education, and AI agents. He will support outreach to AI researchers and educational technology practitioners.

John Stamper is an Associate Professor and MSLE Program Director in Carnegie Mellon University's Human-Computer Interaction Institute. His research spans learning sciences, educational technologies, human-centered AI, and learning analytics. He will support discussions of methods, data, and shared research infrastructure.

Vincent Aleven is a Professor in Carnegie Mellon University's Human-Computer Interaction Institute and Director of the CATS Lab. His research focuses on intelligent tutoring systems, learning analytics, and educational technologies deployed in classroom and online settings. He will support synthesis across learning sciences, AI, and HCI perspectives.

7 Call for Participation

Human-AI systems for learning increasingly require collaboration among communities with different goals, methods, and forms of evidence. HAIL brings together researchers and practitioners from AI, HCI, and the learning sciences, along with educators, designers, students, and other educational stakeholders, to examine how such collaborations can be formed, negotiated, and sustained.

We invite accessible, non-anonymous position papers of one to two pages, excluding references. Submissions may describe lessons from an existing collaboration, a research or practice problem requiring complementary expertise, or a perspective from educational practice. Each paper should identify a concrete experience, challenge, or problem; the communities and stakeholders involved; and one or more questions for workshop discussion. New empirical results are optional. Papers will be reviewed for relevance, specificity, discussion potential, and complementary perspectives.

We expect 40–60 participants and 12–16 accepted position papers. Two consecutive in-person sessions will combine brief thematic presentations, interleaved discussion, and thematic collaboration groups organized around recurring collaboration tensions rather than disciplinary identity or submission category. Participants will compare concrete cases and create collaboration maps capturing relevant expertise and stakeholders, unresolved tensions, and promising practices. All CHI attendees may participate, subject to capacity, with priority for accepted-paper authors.

With authors' permission, accepted position papers and the participant-reviewed workshop synthesis will be posted on the workshop website as non-archival materials. Only the organizer proposal will appear in the CHI Extended Abstracts proceedings. Website: [website]. Submission: [submission-form].

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